V4 Comprehensive Statistical Summary

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CAR V4 STRATIFIED MODEL AUDIT | idata_car_v4.nc
============================================================
WAIC (Deviance): 57926995676.51 | LOO-CV: 15084201.02
Divergences: 0 | Max R-hat: 3.400

PUBLIC HEALTH INTERPRETATION:
- Smoking: -56.00% risk per 5.3% increase
- Male Interaction: -1.00% additional risk
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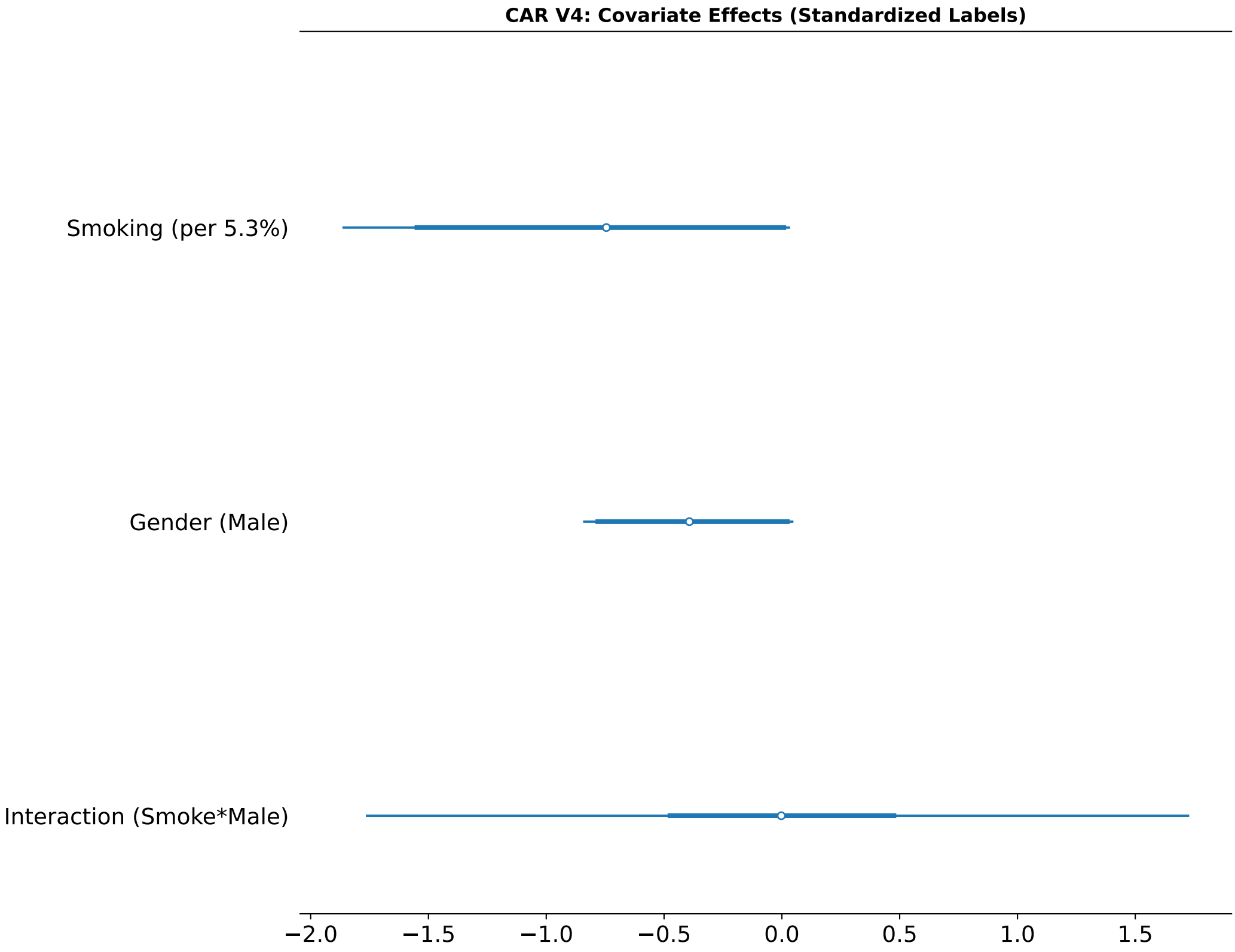
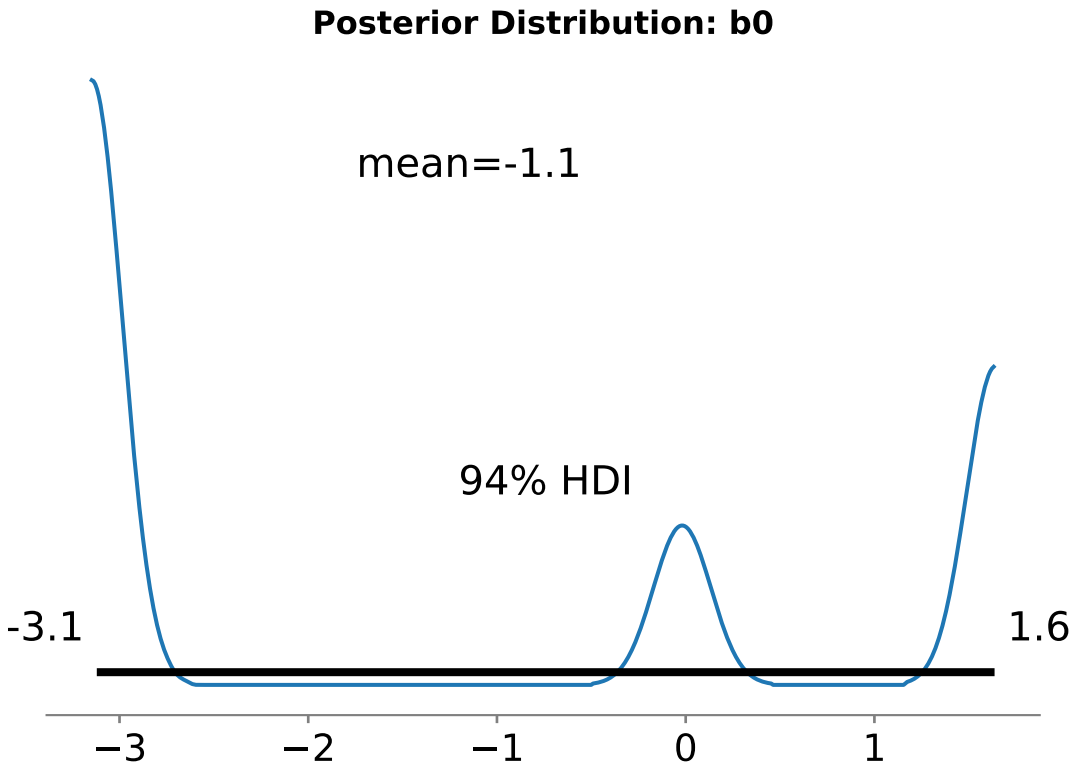
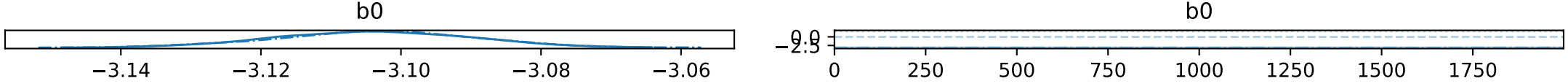
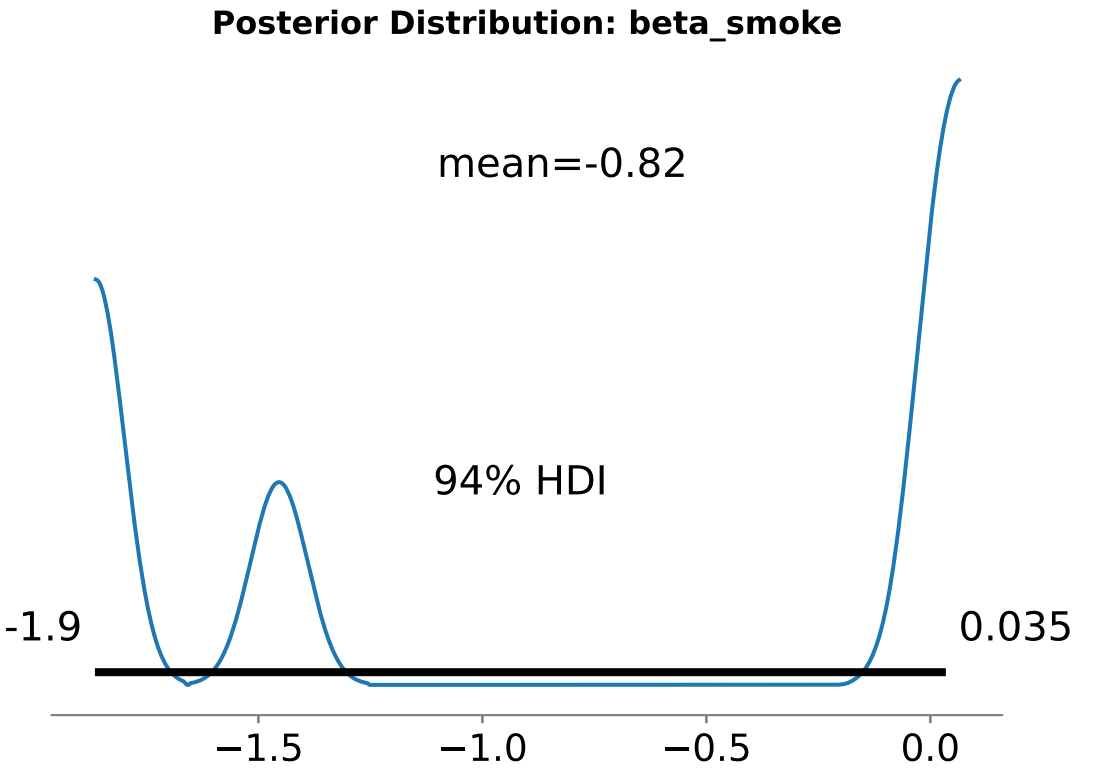
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
b0	-1.148	2.042	-3.120	1.637	1.019	0.283	5.0	108.0	3.37
beta_smoke	-0.821	0.852	-1.865	0.035	0.425	0.071	5.0	4.0	3.40
beta_men	-0.388	0.422	-0.844	0.049	0.210	0.012	5.0	4.0	2.32
beta_interaction	-0.010	1.236	-1.765	1.729	0.617	0.308	4.0	4.0	3.40
sigma	2.282	2.030	0.226	4.457	1.013	0.052	5.0	139.0	3.37
rho	0.552	0.144	0.319	0.742	0.056	0.017	9.0	119.0	1.40


CAR V4: Covariate Effects (Standardized Labels)
Smoking (per 5.3%)
Gender (Male)
Interaction (Smoke*Male)
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

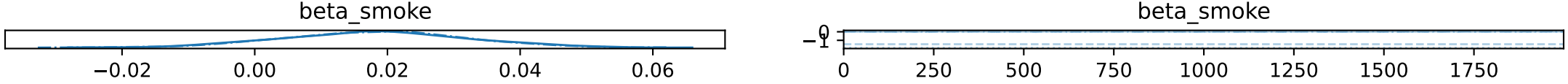

Posterior Distribution: b0
mean=-1.1
94% HDI
-3.1
1.6
−3
−2
−1
0
1

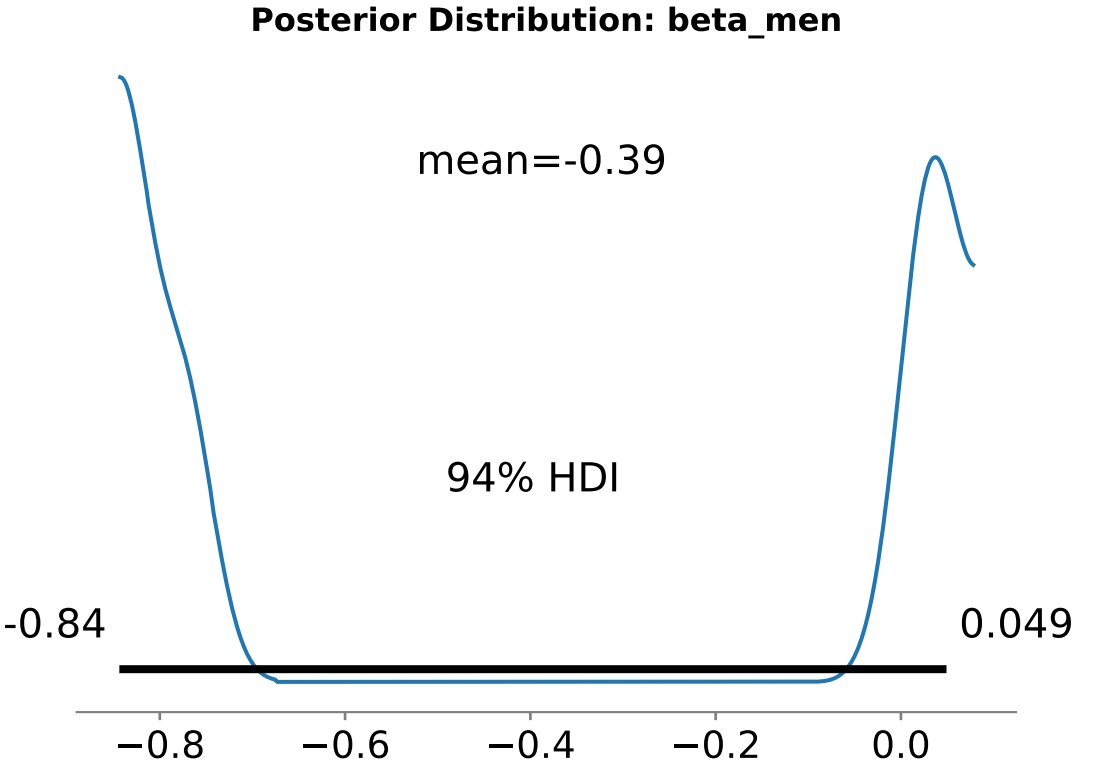
MCMC Chain Trace: b0



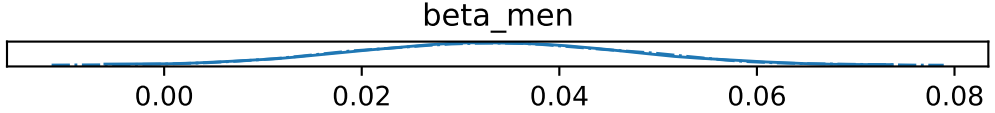

Posterior Distribution: beta_smoke
mean=-0.82
94% HDI
-1.9
0.035
−1.5
−1.0
−0.5
0.0

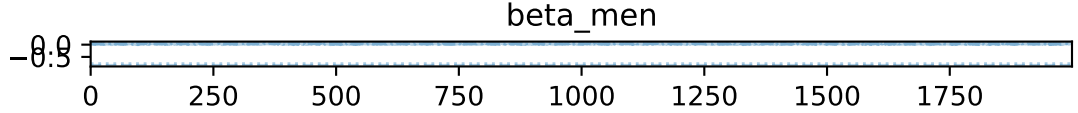
MCMC Chain Trace: beta_smoke

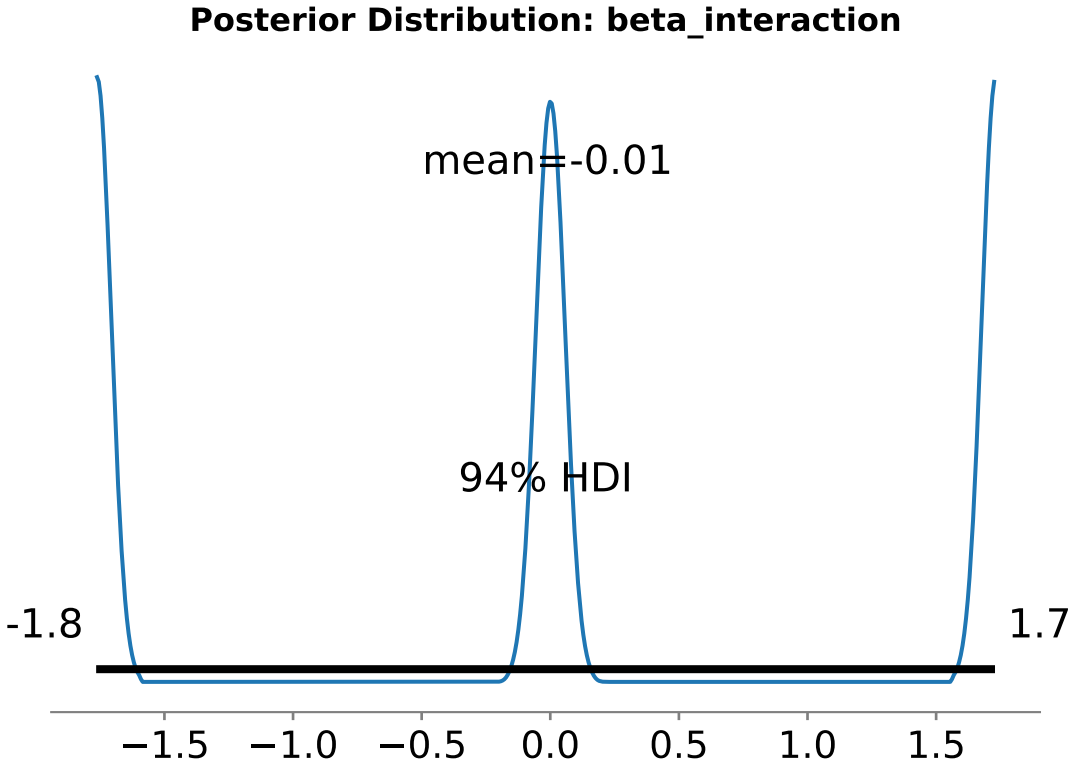



Posterior Distribution: beta_men
mean=-0.39
94% HDI
-0.84
0.049
−0.8
−0.6
−0.4
−0.2
0.0

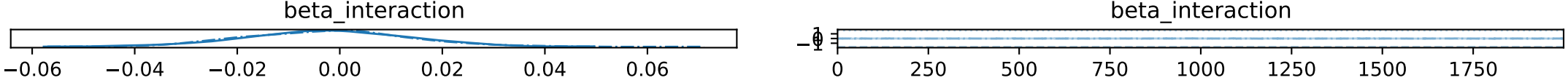
MCMC Chain Trace: beta_men

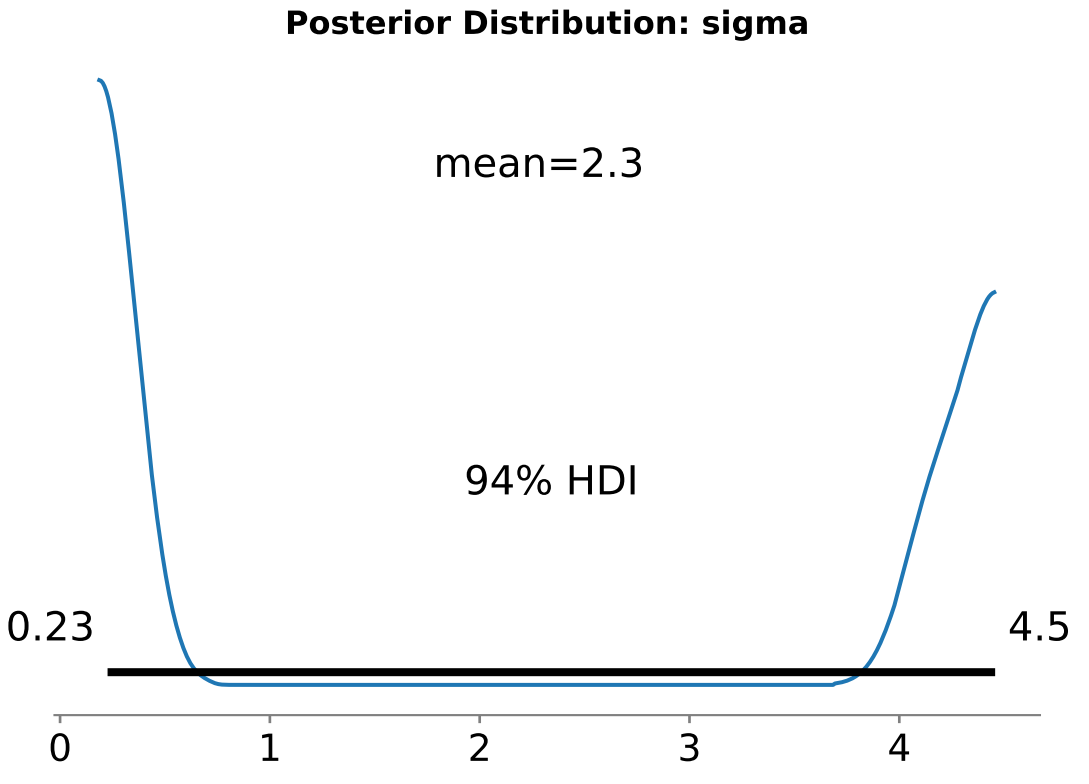




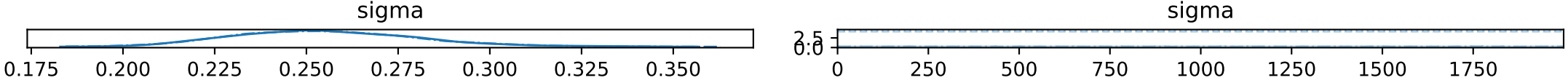

Posterior Distribution: beta_interaction
mean=-0.01
94% HDI
-1.8
1.7
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5

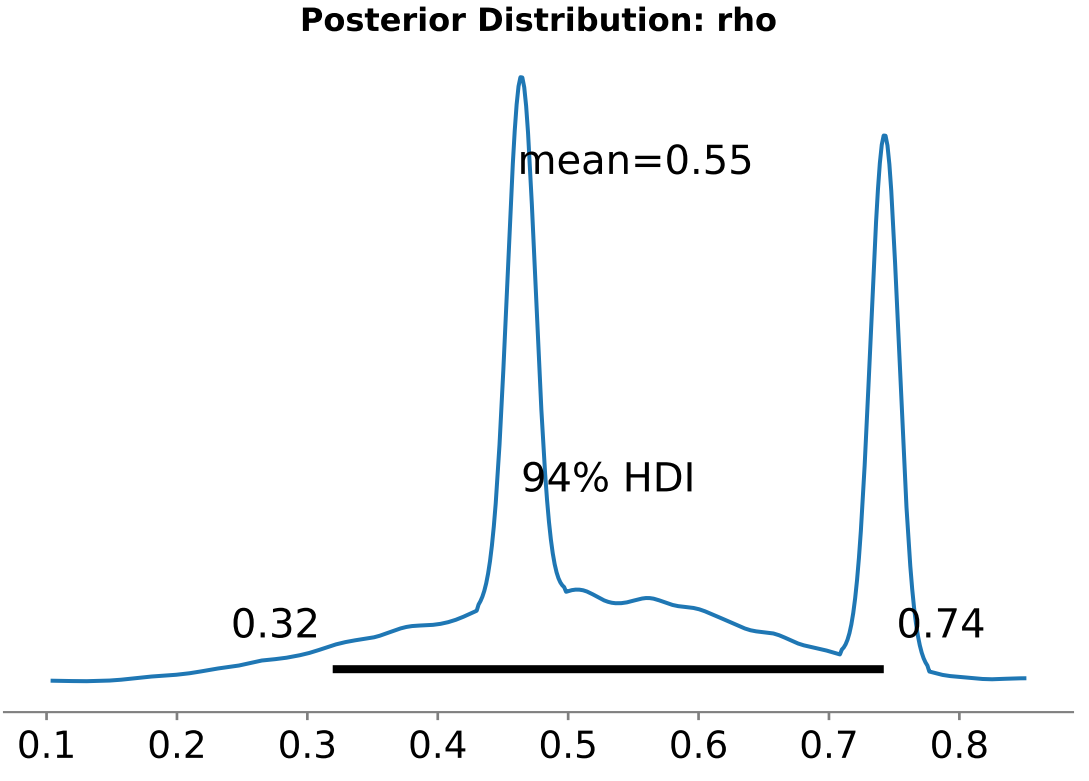
MCMC Chain Trace: beta_interaction




Posterior Distribution: sigma
mean=2.3
94% HDI
0.23
4.5
0
1
2
3
4

MCMC Chain Trace: sigma




Posterior Distribution: rho
mean=0.55
94% HDI
0.32
0.74
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

MCMC Chain Trace: rho